

Memory Map During Boot

Physical Memory Discovery and Categorization

MemUMB

Upper Memory Block

0x000000 - 0x0A0000 (640KB)

ISA devices, BIOS

Reserved

0x0A0000 - 0x1000000

Video memory

MemUPA

Unbacked Physical Address

0x1000000 - 0x100000000 (16MB-4GB)

MMIO mapping space

MemRAM

Usable Physical RAM

0x100000000+ (4GB+)

Available for allocation

Kernel Image

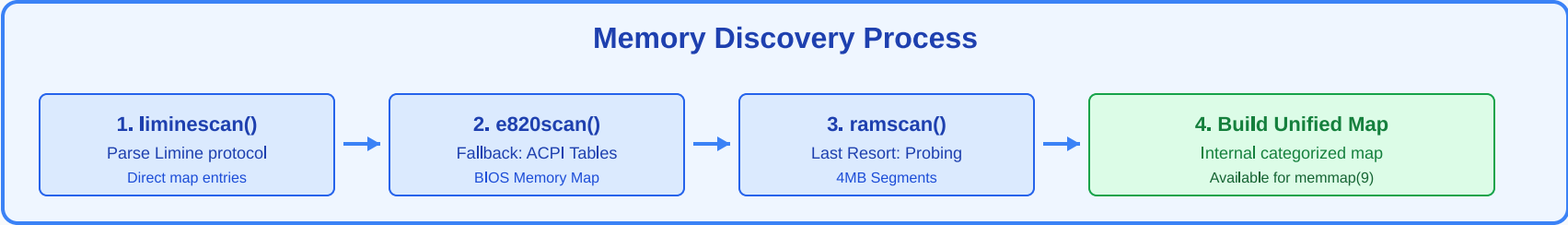
Text + Data + BSS

MemACPI

ACPI Tables

Reclaimable

Bootloader Data



Key Properties

- MemRAM: Allocatable pages for processes
- MemUMB: ISA devices, BIOS ROM space
- MemUPA: Unbacked MMIO register window

Cache Configuration

- MemUMB / MemUPA: Forced UC (Uncacheable)
- MemRAM: Write-Back caching enabled
- MTRR exclusion zones set during bootstrap